

Element D: Design Concept Generation, Analysis, and Selection

Our school produces a lot of paper waste. A large majority of this waste comes from sheets of paper that have only been used or printed on one side. The other side is blank and completely usable, but the paper is getting thrown out anyway. This leads to more cost for the school since they have to pay more money for paper. It also led to more work for our housekeepers, who had to haul all of the extra paper waste out of the building and into dumpsters. The other paper waste comes from the composition notebooks that students use for classes. The cost of these notebooks comes directly out of each teacher's budget. However, in most cases, less than the first third of the notebook is used, and then it is not used again and just gets thrown out, even though the other two-thirds of the notebook have never been used.

Our brainstorming started by individually coming up with a list of possible solutions without fully thinking any of them out. The goal was to write as many ideas as possible; this way, we wouldn't just focus on one idea, and we would be open to lots of other ideas. This was followed by doing the same thing, but as a group, to come up with the largest possible number of ideas. Then we figured out as a group what 5 ideas were realistic and practical. From that, we used a decision matrix, seen below, to choose which of our 5 ideas was best, then started on a prototype.

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Design Requirements	Proposed Solutions					
	Current Solution: Storage box	Idea #1: Notebook with staples & lined paper	Idea #2: Fire starter	Idea #3: Small notebook with staples & blank paper	Idea #4: Nail polish steam machine	Idea #5: Mushy paper tech
Safe process, no sharp objects/ edges, final product is safe to touch - x3	0	0	0	0	-3	0
At least one side blank - x3	0	3	3	3	3	3
Convenient size, A4 to A7 - medium x2	0	0	2	2	0	2
Less than \$2.00 per solution - high x3	0	0	0	0	-3	0
Reuse 90% of scrap paper from misprints & notebooks - low x1	0	1	1	1	1	1
Notebooks with at least 50 sheets of paper - low x1	0	1	1	1	1	1
The paper is connected	0	1	1	1	0	0
together in some way. - low x1						
The teacher will be able to use- high x3	0	3	0	2	3	0
Students will be able to use	0	3	0	1	3	0
Total Score	0	12	8	11	5	7

Scoring

0 = same as current solution

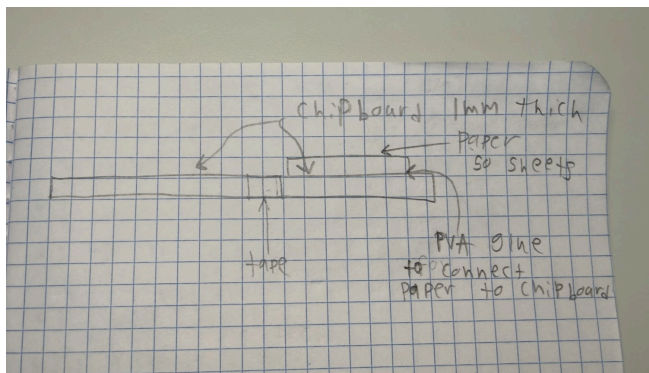
+1 = better than the current solution

-1 = worse than the current solution

Our final solution was to disassemble the used notebooks by cutting the bound edge off and then make new notebooks out of the remaining blank pages. We chose this solution because there were a lot of partly used notebooks with more than half of their pages blank. These notebooks can be made into new notebooks, saving the school money and preventing the blank paper from going to a landfill and being wasted.

Our first step in constructing our solution was to collect used notebooks and sort them by whether they had gridded or lined paper. Then we will cut $\frac{1}{4}$ inch off the notebook on the side that was bound together. We will remove the cover and used pages, and then count out 30 blank pages in a stack and staple them together with three staples on the side. We may have to take paper from more than one notebook to do this. Finally, we will cut chipboard covers to the size of our paper and glue the bottom page of the stack of paper to the bottom of the cover.

Our final solution was to create a notebook out of the unused pages from past notebooks. This new notebook will have 30 sheets with 60 usable pages. It will have a durable chipboard cover to protect the notebook from its environment. The cover was connected to the stack of paper with PVA glue. The stack of paper was held together using 3 staples spaced evenly along the side of the paper.



notebook drawing 4/7/26

